

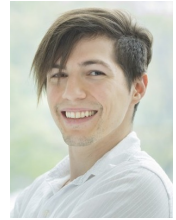
# Marco Antonio Florenzano Mollinetti, Ph.D.

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in Marco Mollinetti

🌐 <http://mollinetti.github.io/>



## Professional Summary

Marco is a Ph.D.-qualified AI leader whose career spans three reinforcing pillars: **fundamental research**, **hands-on engineering**, and **strategic management**. He began as a *researcher*—earning a Ph.D. in Systems Engineering, leading multi-institutional teams, and securing a \$200,000 research grant—evolved into an *engineer* who ships production AI systems (deep-learning architectures at 95% accuracy, autonomous detection pipelines, weather-forecasting integrated into aviation maps), and now operates as a *manager* driving enterprise AI transformation.

As AI Division Manager at the Nohara Group—a Japanese construction conglomerate with over 300 years of history—Marco leads 9+ simultaneous AI projects across Japan, Singapore, and Indonesia. He combines machine learning, optimization, and agentic AI with Lean/Agile practices (A3 thinking, PDCA, Kaizen, Design Sprints, Gemba Walks) to deliver measurable impact: ¥40M in avoided outsourcing costs, up to ¥900M in projected annual revenue uplift, and a 66% reduction in manual specification-entry time. He authors Nohara's AI governance frameworks, runs reskilling and literacy programs, and coaches data-science teams across three countries.

*Available for consulting engagements and selected projects in applied AI, optimization, agentic systems, and AI strategy/governance—remote or on-site, worldwide.*

## Employment History

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| Aug 2025 – . . . .  | <ul style="list-style-type: none"><li>• <b>AI Division Manager</b>, Nohara Group (Japan).<ul style="list-style-type: none"><li>• Leads the AI division within Technology Headquarters: team of 3–5 (including three PhDs), 9+ concurrent projects, and coordination across six+ business units in Japan, Singapore, and Indonesia.</li><li>• <b>Gypsum board allocation &amp; nesting optimization</b><br/>Neuro-symbolic engine (Google OR-Tools CP-SAT + Swin Transformer V2 + Qwen-2.5-32B) with three concurrent engineering streams on a secure, air-gapped cluster. PoC reached 80% yield parity, avoiding ¥40M in outsourced CAD development.</li><li>• <b>AI-powered material quantity takeoff</b><br/>Automated interior material calculation from contractor specifications; lead time cut from five days to under two, with up to ¥900M projected annual revenue uplift and ¥25M saved vs. outsourcing.</li><li>• <b>Hybrid OCR + LLM delivery-sheet pipeline</b><br/>Extracts six mandatory fields from low-quality handwritten faxes; Streamlit dashboard on AWS; 200K-row historical analysis for loss reduction.</li><li>• <b>Agentic AI for legacy estimation logic</b><br/>Copilot Studio agents analyzing VBA macros and BIM plugin workflows; secure local-AI environment for proprietary logic.</li><li>• <b>Cross-border sales AI initiative</b><br/>Copilot agent deployment, SharePoint restructuring, and training for Indonesia and Singapore sales teams.</li><li>• Also delivers AI governance frameworks (Tooling &amp; Engineering Guidelines v1.1, company-wide AI policies, AI Skill Map), Lean/Agile operating rhythm (Gemba Walks, A3, Design Sprints, MVP-first delivery), and AI literacy programs including a 4-week reskilling bootcamp and Nohara AI Academy prototype.</li></ul></li></ul> |
| May 2024 – Aug 2025 | <ul style="list-style-type: none"><li>• <b>AI Team Leader</b>, MTI Ltd. (Japan).<ul style="list-style-type: none"><li>• Diffusion-network manga storyboards (45% productivity, 10% sales); GPT distillation for embedded devices; productized 10-person data-scientist training program; graph-based agentic AI for industrial machinery.</li></ul></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Jan 2022 – May 2024 | <ul style="list-style-type: none"><li>• <b>Senior Data Scientist</b>, MTI Ltd. (Japan)<ul style="list-style-type: none"><li>• Japan–Vietnam–Singapore container-scheduling agent (30% wait-time reduction); rain-forecast system for aviation maps (95% accuracy); three-year Vietnam coaching and training leadership.</li></ul></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Mar 2022 – Feb 2023 | <ul style="list-style-type: none"><li>• <b>Postdoctoral Researcher</b>, CENSIPAM (Brazil) — autonomous oil-spill/vessel detection (UNet + classifiers, +10% accuracy).</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Oct 2019 – Dec 2021 | <ul style="list-style-type: none"><li>• <b>AI Engineer</b>, Plimes Inc. (Japan) — cough/swallow recognition at 95% accuracy with novel framing and wavelet denoising.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Jun 2016 – . . . .  | <ul style="list-style-type: none"><li>• <b>Research Fellow</b>, Lab-TeC (Brazil) — supervised undergrad through doctoral researchers; \$200K CNPq grant for Amazon lightning-detection research (12-person team with UFPA and University of Arizona).</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## Additional Projects

- BelemConverse  Leads a three-person team that developed and fine-tuned a small-scale LLM serving as an AI-powered tour guide for Belém do Pará—host city of the **COP30** global environment conference. The system generates customized tourist routes and recommends restaurants, bars, and lodging in 10+ languages. **Production version released** following successful PoC and development phases.

## Education

- 2018 – 2021  **Ph.D., University of Tsukuba** in Systems Engineering.  
Thesis: *Deterministic Methods for Non-Deterministic Optimization Algorithms*.
- 2016 – 2018  **M.Sc. Computer Science, University of Tsukuba** in Systems Engineering.  
Thesis: *A Bee-Restarting Algorithm for the Nelder-Mead Descent Method*.
- 2011 – 2014  **B.A. Computer Science, University Centre of Pará**.  
Thesis: *Earthquake prediction model using Evolutionary Computation*.

## Selected Research Publications

- 1 B. B. Gatto, M. A. Mollinetti, E. M. dos Santos, A. L. Koerich, and W. S. da Silva Junior, “A novel genetic algorithm approach for discriminative subspace optimization,” in *Brazilian Conference on Intelligent Systems*, Springer, 2024, pp. 64–79.
- 2 B. B. Gatto, E. M. dos Santos, M. A. Molinetti, and K. Fukui, “Multilinear clustering via tensor fukunaga–koontz transform with fisher eigenspectrum regularization,” *Applied Soft Computing*, vol. 113, p. 107 899, 2021.
- 3 M. A. F. Mollinetti, B. B. Gatto, M. T. R. S. Neto, and T. Kuno, “A-dvm: A self-adaptive variable matrix decision variable selection scheme for multimodal problems,” *Entropy*, vol. 22, no. 9, p. 1004, 2020.
- 4 R. L. Pereira et al., “Game theory and social interaction for selection and crossover pressure control in genetic algorithms: An empirical analysis to real-valued constrained optimization,” *IEEE Access*, vol. 8, pp. 144 839–144 865, 2020.
- 5 M. A. F. Mollinetti, D. L. Souza, R. L. Pereira, E. K. K. Yasojima, and O. N. Teixeira, “Abc+ es: Combining artificial bee colony algorithm and evolution strategies on engineering design problems and benchmark functions,” in *Hybrid Intelligent Systems: 15th International Conference HIS 2015 on Hybrid Intelligent Systems, Seoul, South Korea, November 16-18, 2015*, Springer, 2016, pp. 53–66.

## Skills

|                         |                                                                                                                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Coding                  |  Python, R, Java, C++, $\text{\LaTeX}$ , MATLAB, Lua, PHP; Google OR-Tools (CP-SAT), Streamlit, lxml                        |
| Project & Collaboration |  Jira, Git, Trello, Miro, Confluence, Microsoft Loop, SharePoint                                                            |
| Cloud & AI Platforms    |  AWS, Google Cloud, Azure, Microsoft Copilot Studio, Microsoft 365 Copilot                                                  |
| Generative AI           |  Diffusion models (DALL-E, Midjourney, Stable Diffusion, Flux); LLMs (GPT-4, Claude, Mistral, Llama, Qwen local deployment) |
| Machine Learning        |  Transformers, UNets, VAEs; deep Q-learning; hierarchical clustering, PCA; experimental design and regression               |
| Optimization & OR       |  Nonlinear and combinatorial optimization, operations research, constraint programming                                      |

## Skills (continued)

|                         |                                                                                                                                                             |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Domain                  | Construction technology (ConTech), BIM, Japanese building standards (JASS 26), CEDXM                                                                        |
| Lean/Agile              | A3 thinking, PDCA, Kaizen, Design Sprints, Gemba Walks, MVP-driven delivery, concurrent engineering                                                         |
| Leadership & Management | Portfolio management, stakeholder and executive reporting, cross-cultural team leadership, vendor evaluation, Lean/Agile coaching, data-scientist mentoring |
| Languages               | <b>Native:</b> English, Portuguese. <b>Fluent:</b> Japanese (professional daily use; N1 equivalent), Spanish. <b>Intermediate:</b> Italian.                 |

## Miscellaneous Experience

### Awards and Achievements

- 2022 **Project Grant for Research**, National Council of Scientific and Technological Development (CNPq), Brazil.
- 2018 **Best Research Paper**, Brazilian Conference on Computational Intelligence (BRACIS).
- 2014 **First in Class, Class of 2011**, University Centre of Pará.

### Speaking & Invited Presentations

- Nohara Group (internal)** — Recurring keynotes, seminars, and workshops on enterprise AI strategy, applied machine learning, AI literacy, agentic systems, Copilot adoption, Lean/A3 problem-solving, portfolio management, and cross-border team enablement (Japan, Singapore, Indonesia).
- CESUPA** (Centro Universitário do Estado do Pará) — Guest lectures and invited talks on applied AI, optimization, data-science practice, and technical leadership for undergraduate and graduate audiences.

### Certification

- 2023 **AWS Solutions Architect Associate**, Amazon Web Services.
- 2017 **TOEIC Written Test**, Score: 990/990.
- 2015 **TOEFL iBT**, Score: 108/120.
- 2014 **Japanese Language Proficiency Test (JLPT) N3**; professional fluency at N1 equivalent through daily use at Nohara Group.